Chapter 1

1.1 Level 1

Question 1: std::print

What does std::print provide compared to printf or std::cout?

Answer 1: std::print

Type safety, checked at compile time. Can format cutom types (with std::formatter) Uses python formatting Very concise and easy to use.

Question 2: auto

What is the advantage of auto and type deduction in moder C++?

Answer 2: auto

Prevent types mismatches.
Easy use for templates
clean syntax

Question 3: Passing by _

What is the difference between passing by value, reference, and const reference?

Answer 3: auto

value complete copy reference use a pointer to edit the original variable const ref used like a read only of the original variable

Question 4: bindings

What are structured bindings and when would you use the?

Answer 4: auto

Structured binding upacks and object (tuple, array, etc..) into named variables. You have direct access with auto& to update the original variable. Used for readability for complicated things like tuples or maps.

Question 5: Ranges

What are C++23 ranges and views?

Answer 5: auto

Question 6: constexpr

What does constexpr mean and when is it evaluated?

Answer 6: auto

Question 7: Vector vs Arrays

What is the difference between std::vector and raw arrays?

Answer 7: auto

Question 8: string_view

What is std::string_view and why is it more efficient than std :string?

Answer 8

Question 9: ranges

What is the purpose of #include ⋘ranges>

Question 10: lifetime

What is the lifetime of a temporary returned from a function?